Callback

Übungen

Youtube

Implementiere die Methoden in User mit Hilfe von Callbacks, sodass die kommertierte Ausgabe ausgegeben wird.

```java
class Youtube {

  /**
   * Implement the methods in {@link User} by using Callbacks so that the following output is printed.
   */
  public static void main(String[] args) {
    var elona = new User("Elona");
    var anna = new User("Anna");
    var jeff = new User("Jeff");
    var sara = new User("Sara");

    elona.subscribe(anna);
    jeff.subscribe(anna);
    elona.subscribe(jeff);
    jeff.subscribe(elon);
    sara.subscribe(elon);

    elona.upload(new Video("If You Feel Burnt Out Do This"));
    jeff.upload(new Video("The Right Way To Build REST APIs"));
    anna.upload(new Video("Try on haul 2024"));

    /*
    Sara has been notified of a new video 'If You Feel Burnt Out Do This' by Elona
    Jeff has been notified of a new video 'If You Feel Burnt Out Do This' by Elona
    Elona has been notified of a new video 'The Right Way To Build REST APIs' by Jeff
    Elona has been notified of a new video 'Try on haul 2024' by Anna
    Jeff has been notified of a new video 'Try on haul 2024' by Anna
     */
  }
}
```

```java
class User {

  private final String name;
  private final Collection<Video> videos = new ArrayList<>();

  User(String name) {
    this.name = name;
  }

  void upload(Video video) {
    videos.add(video);
    // TODO
  }

  private void notifySubscriberOfNewVideo(User uploader, Video video) {
    System.out.printf("%s has been notified of a new video '%s' by %s%n", name, video, uploader);
  }

  void subscribe(User subscriber) {
    // TODO
  }

  Collection<Video> allUploadedVideos() {
    return Collections.unmodifiableCollection(videos);
  }

  @Override
  public String toString() {
    return name;
  }
}
```

```java
record Video(String title) {

  @Override
  public String toString() {
    return title;
  }
}
```

Youtube: Lösung

```
class User {

   private final String name;
   private final Collection<Video> videos = new ArrayList<>();
   private final Set<User> subscribers = new HashSet<>();

   User(String name) {
      this.name = name;
   }

   void upload(Video video) {
      videos.add(video);
      subscribers.forEach(user -> user.notifySubscriberOfNewVideo(this, video));
   }

   private void notifySubscriberOfNewVideo(User uploader, Video video) {
      System.out.printf("%s has been notified of a new video '%s' by %s%n", name, video, uploader);
   }

   void subscribe(User subscriber) {
      subscriber.subscribers.add(this);
   }

   Collection<Video> allUploadedVideos() {
      return Collections.unmodifiableCollection(videos);
   }

   @Override
   public String toString() {
      return name;
   }
}
```

Call of Shooty: Code

Gegeben ist folgender Code.
Schreibe einen Unit Test mit 100% Testabdeckung für die Klasse **GameEngine**.

```
@FunctionalInterface
interface PlayerEliminated {

   void playerEliminated(Player eliminator, Player eliminatee);
}
```

```
class Player {

   private final String name;
   private int score;

   Player(String name) {
      this.name = name;
   }

   String name() {
      return name;
   }

   int score() {
      return score;
   }

   void incrementScore() {
      score++;
   }

   @Override
   public String toString() {
      return "Player{name='" + name + '\'' + ", score=" + score + '}';
   }
}
```

```
import java.util.*;

class GameEngine {

   private static final int PLAYERS_TO_CREATE = 10;
   private static final Random RANDOM = new Random();

   private final List<Player> players = new ArrayList<>();
   private final Collection<PlayerEliminated> playerEliminatedSubscribers = new ArrayList<>();

   GameEngine() { createPlayers(); }

   private void createPlayers() {
      for (int i = 0; i < PLAYERS_TO_CREATE; i++) {
         String defaultName = "Player" + i;
         players.add(new Player(defaultName));
      }
   }

   void subscribePlayerEliminated(PlayerEliminated subscriber) {
      playerEliminatedSubscribers.add(subscriber);
   }
   void notifyPlayerEliminated(Player eliminator, Player eliminatee) {
      playerEliminatedSubscribers.forEach(subscriber -> subscriber.playerEliminated(eliminator, eliminatee));
   }

   void run() {
      Player eliminator = getRandomPlayer();
      Player eliminatee = getRandomPlayer();
      if (eliminator != eliminatee) {
         eliminator.incrementScore();
      }
      notifyPlayerEliminated(eliminator, eliminatee); // eliminator has just shot eliminatee
   }

   private Player getRandomPlayer() { return players.get(randomPlayerIndex()); }

   private int randomPlayerIndex() { return RANDOM.nextInt(0, players.size()); }
}
```

Call of Shooty: Lösung

```java
import org.junit.jupiter.api.*;
import static org.junit.jupiter.api.Assertions.*;

class GameEngineTest {

    private GameEngine gameEngine;

    @BeforeEach
    void setUp() {
        gameEngine = new GameEngine();
    }

    @Test
    void publishSubscribePlayerEliminated() { // this test case is a bit hacky (difficult to test).
        // Arrange
        boolean[] wasCalled = new boolean[1];

        // Act
        gameEngine.subscribePlayerEliminated((_, _) -> {
            wasCalled[0] = true; // not pure, but necessary for the test.
        });
        gameEngine.run();

        // Assert
        assertTrue(wasCalled[0]);
    }

    @RepeatedTest(10)
    void scored() { // due to randomness, repeat this test case ten times
        // Assert
        gameEngine.subscribePlayerEliminated((eliminator, eliminatee) -> {
            if (eliminator != eliminatee) { // a bit tricky to test due to randomness.
                assertEquals(1, eliminator.score());
                assertEquals(0, eliminatee.score());
            } else {
                assertEquals(0, eliminator.score());
            }
        });

        // Act
        gameEngine.run();
    }
}
```

Student Modified Callback

Gegeben ist die Klasse **Student** inklusive **Unit Test** (siehe nächste Folie).

Ergänze die Klasse Student, sodass man **beliebig viele Callbacks** hinzufügen kann, die aufgerufen werden, sobald sich irgendeines der Attribute in Student geändert hat (subscribeOnModified).

Die hinzugefügten Callbacks werden **evaluiert**, **sobald** sich eines der **Attribute** in Student **ändert** – also sobald einer der Setter aufgerufen wird. Der Callback soll den **geänderten Student** als Parameter erhalten.

Teste deine **Callbacks**, indem du den **Unit Test erweiterst**.

```
// Beispiel:
public static void main(String[] args) {
    var student = new Student(1, "Elon");
    student.subscribeOnModified(modifiedStudent -> System.out.println(modifiedStudent));
    System.out.println(student);
    student.setId(2); // evaluate callback
    student.setName("Elon Musk"); // evaluate callback
}
```

Student Change: Code

```
class Student {

   private int id;
   private String name;

   Student(int id, String name) {
      this.id = id;
      this.name = name;
   }

   int id() {
      return id;
   }

   void setId(int id) {
      this.id = id;
   }

   String name() {
      return name;
   }

   void setName(String name) {
      this.name = name;
   }

   @Override
   public String toString() {
      return "Student{id=" + id + ", name='" + name + '\'' + '}';
   }
}
```

```
class StudentTest {

   private Student student;

   @BeforeEach
   void setUp() {
      student = new Student(1, "Elon");
   }

   @Test
   void getSetId() {
      student.setId(2);

      assertEquals(2, student.id());
   }

   @Test
   void getSetName() {
      student.setName("Elon Musk");

      assertEquals("Elon Musk", student.name());
   }

   @Test
   void toStringWithIdAndName() {
      var actual = student.toString();

      assertEquals("Student{id=1, name='Elon'}", actual);
   }
}
```

Student Change: Lösung

```java
class Student {

   private int id;
   private String name;

   // Alternative: custom functional interface with one parameter and void return value.
   private final Collection<Consumer<Student>> onModifiedSubscribers = new LinkedList<>();

   Student(int id, String name) {
      this.id = id;
      this.name = name;
   }

   void subscribeOnModified(Consumer<Student> subscriber) { // Alternative: addOnModifiedSubscriber
      this.onModifiedSubscribers.add(subscriber);
   }

   private void publishOnModified() { // Alternative: notifyOnModifiedSubscribers()
      onModifiedSubscribers.forEach(subscriber -> subscriber.accept(this));
   }

   int id() { return id; }
   String name() { return name; }

   void setId(int id) {
      this.id = id;
      publishOnModified();
   }

   void setName(String name) {
      this.name = name;
      publishOnModified();
   }

   @Override
   public String toString() {
      return "Student{id=" + id + ", name='" + name + '\'' + '}';
   }
}
```

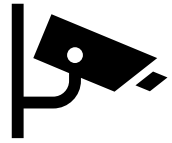
```java
import org.junit.jupiter.api.*;
import static org.junit.jupiter.api.Assertions.assertEquals;

class StudentTest {
   private Student student;

   @BeforeEach
   void setUp() { student = new Student(1, "Elon"); }

   @Test
   void getSetId() {
      student.setId(2);
      assertEquals(2, student.id());
   }
   @Test
   void getSetName() {
      student.setName("Elon Musk");
      assertEquals("Elon Musk", student.name());
   }
   @Test
   void toStringWithIdAndName() {
      var actual = student.toString();
      assertEquals("Student{id=1, name='Elon'}", actual);
   }
   @Test
   void idModified() {
      // Assert
      student.subscribeOnModified(modifiedStudent -> {
         assertEquals(2, modifiedStudent.id());
         assertEquals("Elon", modifiedStudent.name());
      });
      // Act
      student.setId(2);
   }
   @Test
   void nameModified() {
      // Assert
      student.subscribeOnModified(modifiedStudent -> {
         assertEquals(1, modifiedStudent.id());
         assertEquals("Elon Musk", modifiedStudent.name());
      });
      // Act
      student.setName("Elon Musk");
   }
}
```

Big Brother



Gegeben sind folgende Klassen. Vervollständige den Code an den mit // TODO kommentierten Stellen, sodass der unveränderte Unit Test erfolgreich besteht.

```
import java.util.LinkedList;
import java.util.List;

class BigBrother {

   private final List<String> log = new LinkedList<>();

   BigBrother add(Person person) {
      // TODO
   }

   boolean logContains(String line) {
      return log.contains(line);
   }
}
```

```
@FunctionalInterface
interface NameChangedCallback {

   void namedChanged(String oldName, String newName);
}
```

```
import java.util.LinkedList;
import java.util.List;

class Person {

   // TODO

   private String name;

   Person(String name) {
      this.name = name;
   }

   String name() {
      return name;
   }

   void setName(String name) {
      // TODO
   }

   void subscribeNameChanged(NameChangedCallback subscriber) {
     // TODO
   }
}
```

```
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

class BigBrotherTest {

   @Test
   void subscribeAndNotifyOnNameChanged() {
      Person elon = new Person("Elon");
      Person sara = new Person("Sara");

      BigBrother bigBrother = new BigBrother()
         .add(elon)
         .add(sara);

      elon.setName("Musk");
      assertTrue(bigBrother.logContains("Elon changed name to Musk"));

      sara.setName("Lara");
      assertTrue(bigBrother.logContains("Sara changed name to Lara"));
   }

   @Test
   void notNotifyIfNotSubscribedOnNameChanged() {
      Person elon = new Person("Elon");

      BigBrother bigBrother = new BigBrother();

      elon.setName("Musk");
      assertFalse(bigBrother.logContains("Elon changed name to Musk"));
   }
}
```

Big Brother: Lösung

```java
import java.util.LinkedList;
import java.util.List;

class BigBrother {

   private final List<String> log = new LinkedList<>();

   BigBrother add(Person person) {
      person.subscribeNameChanged((oldName, newName) -> {
         String message = "%s changed name to %s".formatted(oldName, newName);
         log.add(message); // not pure lambda
      });
      return this;
   }

   boolean logContains(String line) {
      return log.contains(line);
   }
}
```

```java
import java.util.LinkedList;
import java.util.List;

class Person {

   private final List<NameChangedCallback> nameChangedSubscribers = new LinkedList<>();

   private String name;

   Person(String name) {
      this.name = name;
   }

   String name() {
      return name;
   }

   void setName(String name) {
      String oldName = this.name;
      this.name = name;
      nameChangedSubscribers.forEach(subscriber -> subscriber.namedChanged(oldName, name));
   }

   void subscribeNameChanged(NameChangedCallback subscriber) {
      nameChangedSubscribers.add(subscriber);
   }
}
```